CONTROLLER

WATCHDOG + GATING

DRIVE

SUPPLY

U2
ESP32
CHIP_PU
GPIO43/U0TXD
GPIO44/U0RXD
GPIO15/ADC2_CH4/32K_P
GPIO16/ADC2_CH5/32K_N
GPIO20/USB_D+
GPIO19/USB_D-
GPIO39/MTCK
GPIO40/MTDO
GPIO41/MTDI
GPIO42/MTMS
GPIO0
GPIO1/ADC1_CH0
GPIO2/ADC1_CH1
GPIO3/ADC1_CH2
GPIO4/ADC1_CH3
GPIO5/ADC1_CH4
GPIO6/ADC1_CH5
GPIO7/ADC1_CH6
GPIO8/ADC1_CH7
GPIO9/ADC1_CH8
GPIO10/ADC1_CH9
GPIO11/ADC2_CH0
GPIO12/ADC2_CH1
GPIO13/ADC2_CH2
GPIO14/ADC2_CH3
GPIO17/ADC2_CH6
GPIO18/ADC2_CH7
GPIO21
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
GPIO47
GPIO48
ESP32-S3-DevKitC
+5V
3V3
GND
ISENSE
SDA
SCL
R13 100k
R15 100k
R11 4k7
R14 4k7
J3
to handset, mates with J4
handset sees 3V3 only — 29V never reaches it
C3 2n2
R3 100k
U3A 74HC123 250ms
U3C 74HC123 250ms
RCext
Cext
WDOG
U6A 74HC08
U6B 74HC08
U6C 74HC08
U6E 74HC08
R1 10k
R2 10k
R6 10k
R7 1k
Q4 BC337
Q5 BC327
hold drive lines safe before boot
VSW
SB560 D5
Q1 IRLB8721
MRET
Q2 2N7002
Q3 2N7002
K1
energized = UP (raises desk)
K2
energized = DOWN (lowers desk)
D2
D6
flyback
DRV_UP
U5 ACS724xLCTR-05AB
VCC
VIOUT
F.FILTER
IP+
IP-
GND
R4 10k
R5 20k
J2
to motor housing, mates with J1
D1 33V bidir
C1 220u 50V
U1 LM2596HV 29V-5V
VIN
FB
ON/OFF
GND
OUT
C2 220u 10V

LEG (unmodified)

D3
allows down
SW3
top limit
M1
895 winding
SW4
bottom limit
D4
allows up
J1
to motor housing, mates with J2
PSU1
+Vout 1
-Vout 1
+Vout 2
-Vout 2
AC/L
AC/N
29V 1.8A 52W
L
N

HANDSET (unmodified)

SW1
DOWN rocker
SW2
UP rocker
J4
to handset, mates with J3

SIMPLE IN-LINE DRIVE. The box sits in the 4-core cable and owns the winding permanently. Reverting is physical: motor — box — handset becomes motor — handset.

The handset is no longer in the motor circuit. It is fed 3V3 and its rockers are read as dry contacts — high at rest through the OEM NC contact, low when pressed. Because 29V can never appear on those conductors, no clamp diodes are needed.

POLARITY (measured on the 4-pin Mini-Fit Jr, circuits numbered on the housing):
circuit 1 = motor circuit 3 = supply +
circuit 2 = motor circuit 4 = supply −
circuit 2 positive, circuit 1 negative = UP. Reverse = DOWN.
MOT_A is defined as the conductor that must be POSITIVE to raise the desk (circuit 2). MOT_B is circuit 1.

K1 / K2 reproduce the OEM rest behaviour:
both off -> both winding ends on VSW = BRAKE
K1 on -> MOT_B to return, MOT_A stays positive = UP
K2 on -> MOT_A to return, MOT_B stays positive = DOWN
No combination of relay states can short the supply; there is no shoot-through failure mode.

SEQUENCING: relays change state only with Q1 off and current decayed (~50ms). On a watchdog trip Q1 turns off in microseconds while the relays take milliseconds to release, so the contacts are cold-switched even in a fault. Loss of drive is coast, which is safe because the spindle is self-locking and holds the load unpowered.

U5 sits in the motor leg so it reads current during freewheel as well as during the PWM on-time, which a low-side shunt would not. ISENSE must be on ADC1 (ADC2 dies with WiFi). R1 / R2 / R6 must sit on non-strapping GPIOs.

Box owns the winding; unplug it to revert

Sheet: /	
File: desk.kicad_sch	
Title: Slangerup desk retrofit — in-line drive	
Size: A1	Date:
KiCad E.D.A. 10.0.5	Rev:
	Id: 1/1